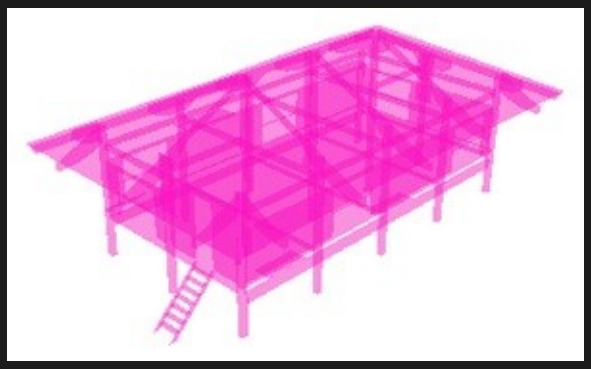
# Modeling in 3DS MAX 5.1

Exporting *.bes Files
Creating Collicion Objects
Creating the Effect Spheres
Creating the Level of Detail (LOD) Structure
User Defined Object Properties
Using PteroMaterial

# Creating Collision Objects



PteroEngine II distinguishes three collision object shape types:

1. automatic
2. defined
3. auxiliary

## Automatic collisions

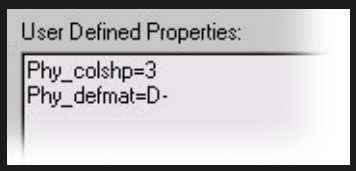
These collisions are created when Phy_colshp and Phy_misshp are set has a collision material assigned.

After the engine loads such model, it computes collision shapes for playe bullets.

If PteroMat has a collision material assigned, both created shapes are graphic shape. Here, it is not necessary to set Phy_colshp or Phy_

For small objects, such as glasses or bottles, we recommend to set to 3. The computation of this collision shape is much faster than If Phy_colshp or Phy_misshp are set to 1 or 3 it is nece

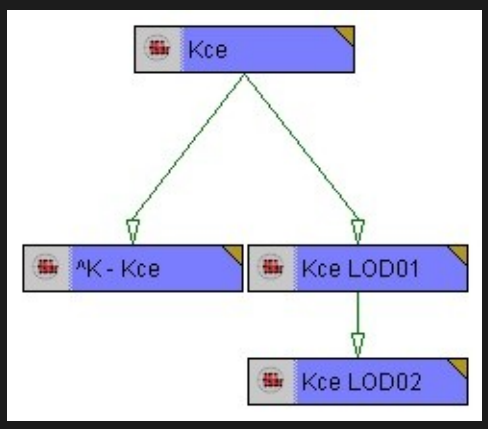
Example:



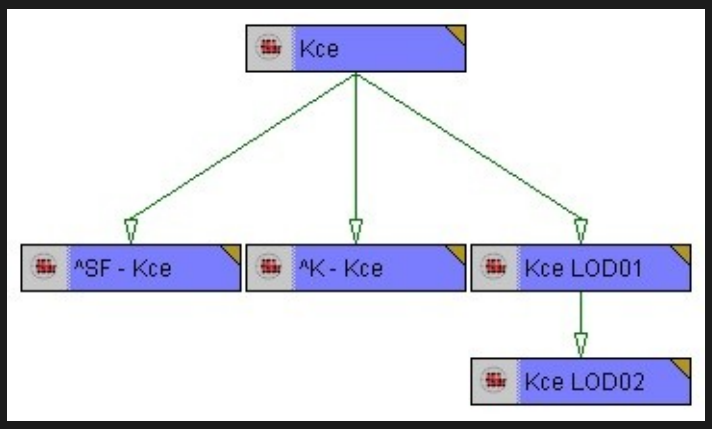

## Defined collisions

Object names of defined collision shapes must begin with ^K, ^SF or It is possible to combine automatic and defined collisions. For instance, defined collision shape for player, link the ^K object to the main bullets will be then automatically created by the engine (but you have

in PteroMat).




Collisions are created for the main objects only, not for its LoD objects.




Apply the collision materials to them using PteroMat.

We recommend to create defined collisions for player. This improves of the game quite a bit.

## Auxiliary collisions

The object names of auxiliary collision shapes must beginwith the '!' mark).

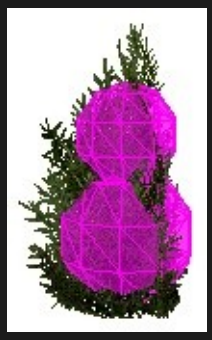
Apply the collision materials to them using PteroMat.

## Naming conventions for defined collision shapes

| | |
|---|---|
| ^K + text | shape for player |
| | |

| | |
|---|---|
| ^SF<br>+ | shape for bullets<br>text |
| ^SK<br>+ | for dummy only: defines sphere contained in dummy<br>text |
| !<br>+ | auxiliary collision<br>text |

# Creating the Effect Spheres



Effect spheres serveas a special collision system. They are used most spheres makeit possible to determine how mucha plant or a shrub and other collision properties.
We recommend that you use this system, because it is muchfaster system.

An effectsphere is created as a sphere within a dummy objec
It cannot be scaled non-uniformly.
The dummy does not need to be linked to anything.

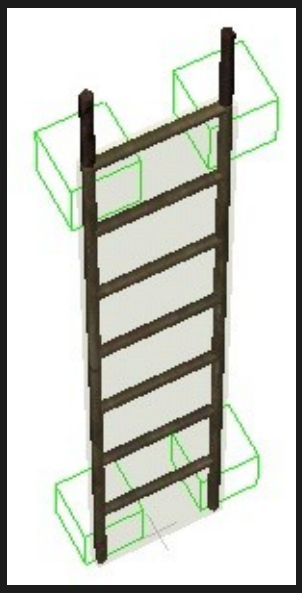
Effect sphere is created using a Dummy object, which name

@[material][slowdown]- + text

| material | two capital letters for material type |
|---|---|
| slowdown | number between 0-9<br>0 = no slowdown<br>9 = almost stops the player |

For example, if we use a collision material JL (leaves), and set slow
@JL3-

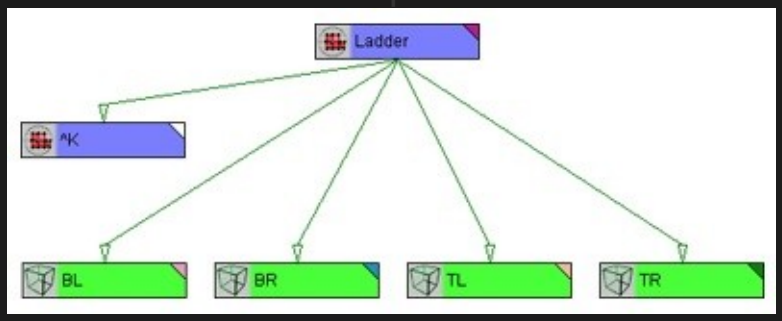
# Creating ladders



A ladder object consists of the main model, simplified collision mode and four dummy objects (BL, BR, TL, TR).

The main object has PteroMat defined with collision material, which serve for bullets.
The collision object for characters (^K) is a bounding box of a main end, the collision object to plane must be aligned with the ladder's last

This is an example of how the auxiliary objects should be linked

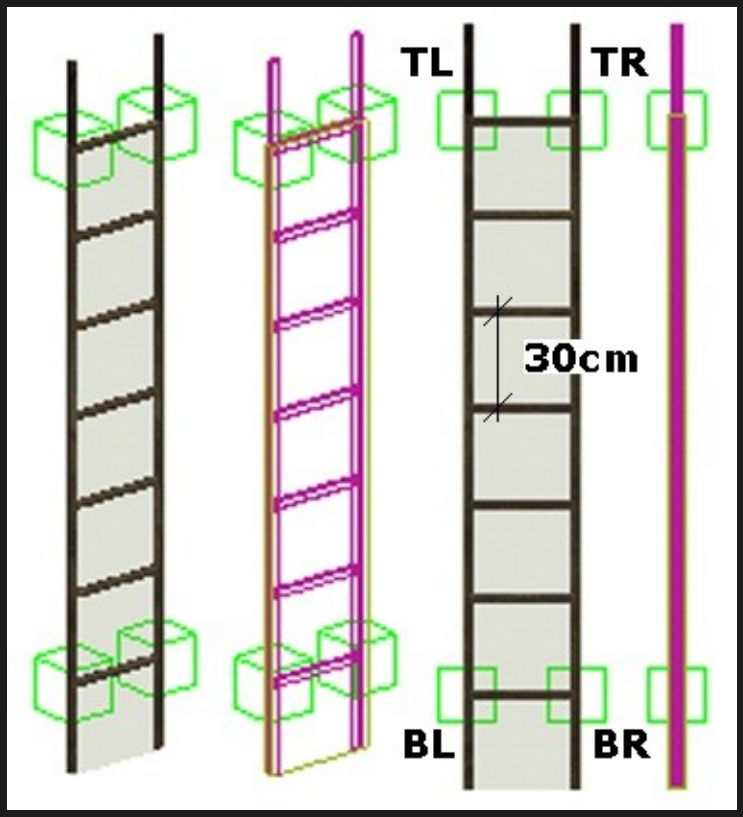



The ladder steps must be 30 cm from each other.

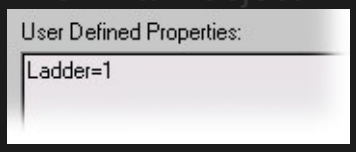
The auxiliary dummy objects determine the position where animation begi and ends (on trhe last step).
These dummy objects must be named according to their position on front view:

| | |
|---|---|
| BL | lower left |
| BR | lower right |
| TL | upper left |
| TR | upper right |

In this case, the ladder is meant to be approached from the front




The main object must have this parameter set:

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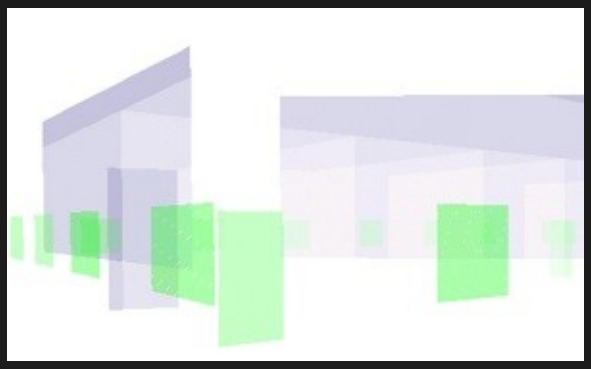
**Creating Collision Objects**

# Creating sectors, portals and occluders

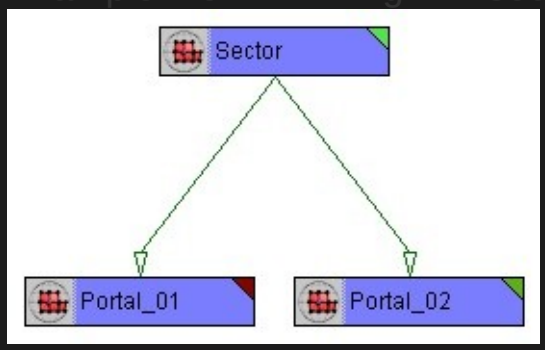


## Creating sectors

Sector is a limited area which can have specific properties related

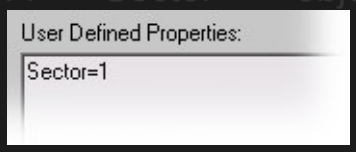
It is defined by a geometric shape, normal vectors of which doesn't need to be convex.
It can contain any number of portals and ocluders. Portals must be

Example of linking sector and portals in schematic view:




Objects associated with the sector are drawn only when some part of secto visible. This means that either at least one of its portals must be inside the sector.

A Sector object must contain proper values and settings in the




## Creating portals

Portal is a shape consisting of triangles, which must be precisely
Its normal vectors must point inside and it must be linked to the

A Portal object must contain proper values and settings in the

User Defined Properties:

Portal=30.0,15.0,10.0,0,4,79

## Creating occluders

Occluder is any shape consisting of triangles. It doesn't have to be can be linked to d.o. It causes that objects with occluding design are not drawn. Portals behind an occluder are not drawn either.

An occluder object must contain proper values and settings in the

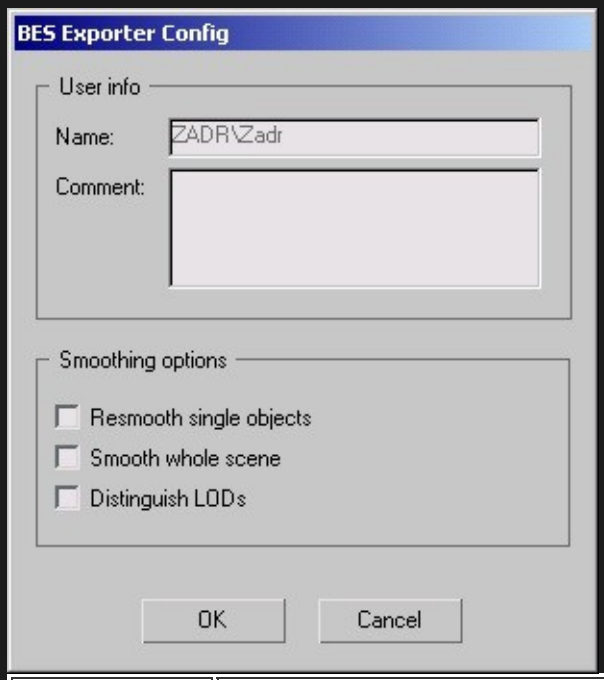
User Defined Properties:

Ocluder=1

# Exporting *.bes Files

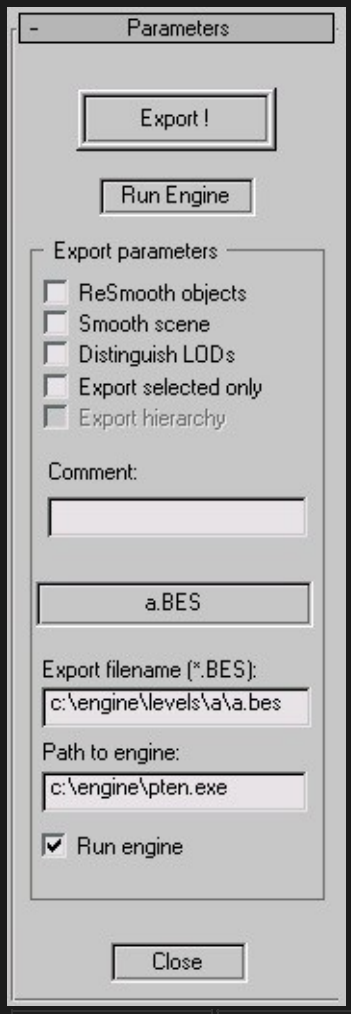
Before you beginwith export, it is necessary to convert all objects perform the "Reset XForm" action. "Reset XForm" is found in As this is a non-reversible action, we recommend to save the scen

| | | | | | | |
|---|---|---|---|---|---|---|
| **menu** | **:** | **file** | **/** | **export...** | **/** | **bes** |
| **menu** | **:** | **file** | **/** | **export selected...** | **/** | **bes (*.bes)** |




| | |
|---|---|
| Name | name of the computer and user - cannot be changed |
| Comment | a comment that can be read in the editor in the Databas |
| Resmooth single objects | re-calculation of neighboring normals - not necessary to us |
| Smooth whole scene | on/off switch for the Smooth parameter in the User Defined |
| Distinguish LODs | this is used for character export when 'Smooth whole scene' |

| | | | | | | |
|---|---|---|---|---|---|---|
| **panel** | **:** | **utilities** | **/** | **more...** | **/** | **BES export** |



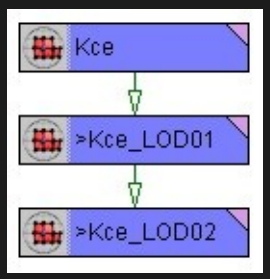

| | |
|---|---|
| Export! | runs the export routine |
| Run Engine | launches PteroEngine - correct path to it must be set |
| Resmooth objects | re-calculation of neighboring normals - not necessary to use |
| Smooth scene | on/off switch for the Smooth parameter in the User Defined |
| Distinguish LODs | this is used for character export when 'Smooth whole scene' |
| Export selected only | exports only selected objects |
| Comment | a comment that can be read in the editor in the Database |
| Export filename | entire path with the *.bes name included |

| | |
|---|---|
| Path to engine | entire path to PteroEngine II |
| Run engine | this switch allows for automatic engine launch and opening object in it |

# Creating the Level of Detail (LOD) Structure

Level of Detail (LOD) is used for decreasing the amount of graphic speeds up rendering of the game.

The auxiliary objects that function as LODs to the main object must according to the graph below which shows a main object with

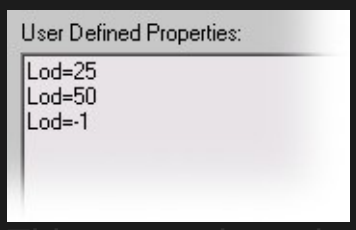



The name of each LOD object must begin with the '>' character.
The position of LOD objects in relation to the main object is deter
We recommend to use 1-3 LODs per object.

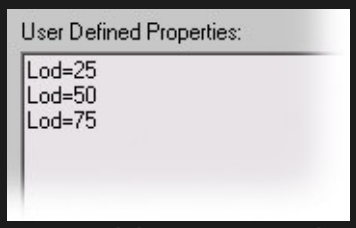
The main object must have the Lod (or LastLodAlpha) parameters assigned.
The number of Lod parameters is the number of LOD objects plus

When setting the Level of Detail, it is also possible to use the

Examples of syntax:




This example shows a case, where there is a main object with two switched on when viewed from 25m distance (and further), the second one
The last Lod parameter is set to -1 which means that the last LOD within the scene's visibility range.




In this example, the last Lod parameter is set to 75. This means that drawn when further than 75m.

User Defined Properties:

Lod=25
Lod=50
Lod=75
LastLodAlpha=1

Here, we have added a parameter named LastLodAlpha. This means that disappear gradually by becoming more increasingly transparent. Again, it drawn entirely when further than 75m.

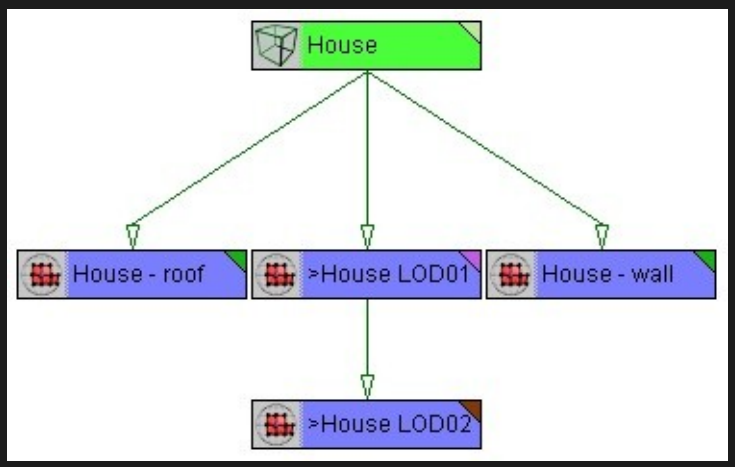
---

**AlphaMult, ClipDist, Dol, DolNoAlpha**
**LastLodAlpha, Lod**
**Dummy object in LOD Structure**

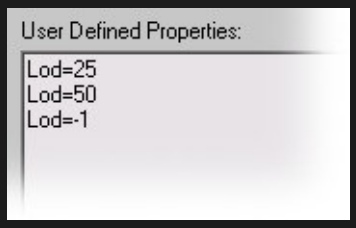
# Dummy object in LOD Structure

A Dummy object can be added to the LOD structure in order to LODs of more mutually independent objects at the same moment.

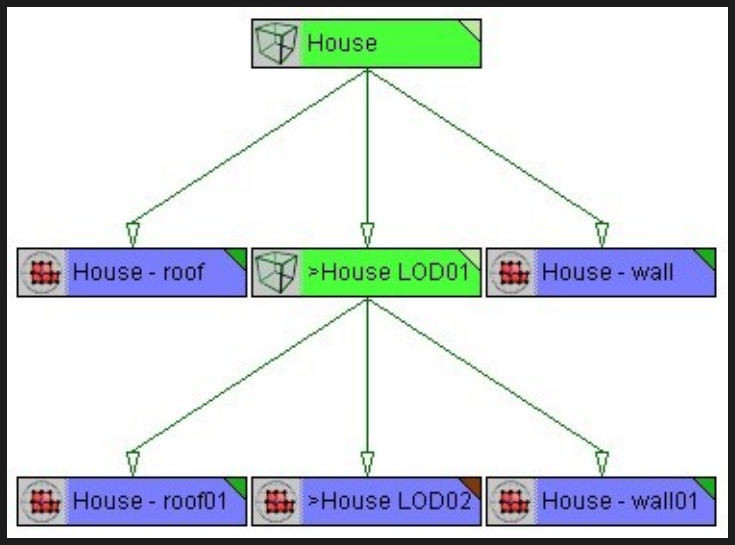
The main objects must be linked to a Dummy as shown below. this dummy, too.




This example shows a Dummy named "House", to which graphic "House - wall" are linked. In the User defined properties of the Dum definition:




This definition would normally belong to the main object of a regul The objects named '>House LOD01' and '>House LOD02' are LOD objec defined distance switched to objects without the '>' character in the




This example shows using a Dummy as both the main and the LOD objec
LOD can act as a parent to a number of graphical objects.

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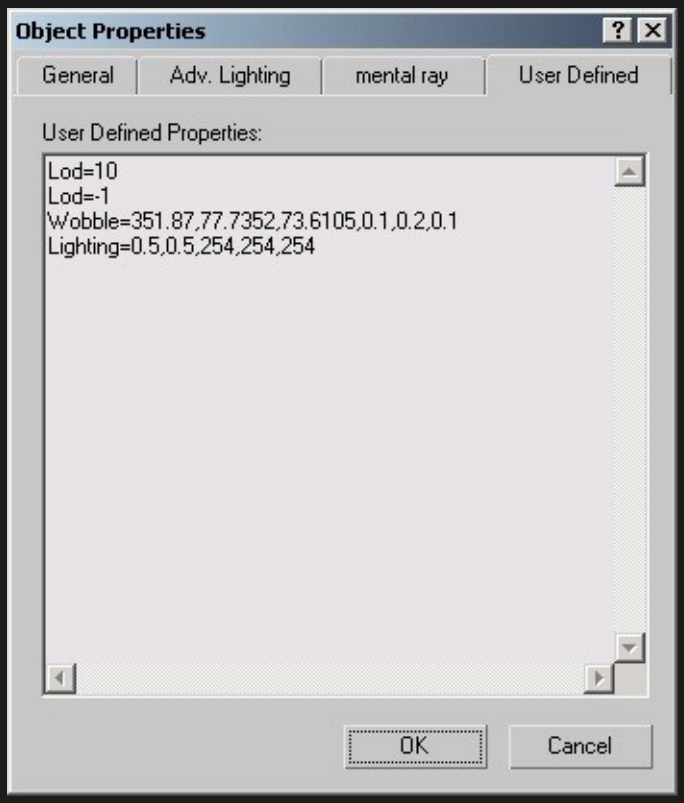
# User Defined Object Properties

**Quad menu : properties / user defined**




These parameters servefor setting an object's properties in editor set parameters of lighting, physics etc.

---

**AlphaMult, ClipDist, Dol, DolNoAlpha**
**Collisions**
**LastLodAlpha, Lod**
**Lighting, Smooth, TexMove, TexPing, Wobble**
**Ocluder, Portal, Sector**
**Physics**

# AlphaMult, ClipDist, Dol, DolNoAlpha - MAX

**Quad menu : properties / user defined**

## AlphaMult= val

User Defined Properties:
AlphaMult=1

- multiplier of transparency distance

| | |
|---|---|
| val | range = 0 až 1<br>default = 0.8 |

## ClipDist= dist

User Defined Properties:
ClipDist=47

- sets the object's distance from the camera in meters where it its subordinate hierarchy
- we recommend to use this parameter for display/scene optimization
- it is possible to use a Dummy for optimization purposes even editor.

## Dol= dist

User Defined Properties:
Dol=5

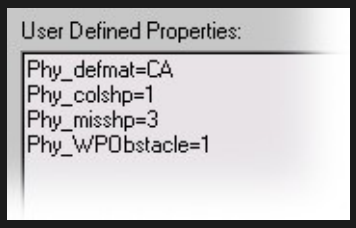
- this parameter sets the max. distance between object and camera, in fully visible

## DolNoAlpha= 1

User Defined Properties:
DolNoAlpha=1

- the switch for disappearing of an object with the Dol parameter

**Quad menu : properties / user defined**




## Phy_colshp= val

- collision shape type for player, AI and physics

| | |
|---|---|
| 0 | a linked shape - automatically set, if the export finds a |
| 1 | creates a bounding box around object |
| 2 | creates collisions from the graphic shape: automatically set, when c set in PteroMat and 0 has not been set yet |
| 3 | creates a bounding sphere around the object |

## Phy_misshp= val

- collision shape type for bullets

| | |
|---|---|
| 0 | a linked shape - automatically set, if the export finds a |
| 1 | creates a bounding box around object |
| 2 | creates collisions from the graphic shape: automatically set, when c set in PteroMat and 0 has not been set yet |
| 3 | creates a bounding sphere around the object |

## Phy_defmat= mat

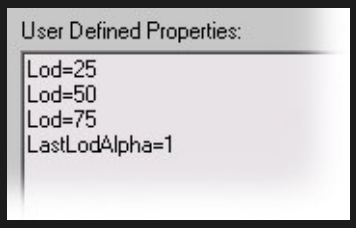
- collision material type
- it is necessary to fillthis in only when Phy_colshp and Phy_missh

## Phy_WPObstacle= 1

- switch for cancellation of waypoint connection with an object

# LastLodAlpha, Lod - MAX

**Quad menu : properties / user defined**




## LastLodAlpha= 1

- switches gradual disappearance of last LoD through transparency.

## Lod=dist

- this enables LoD switching based on the distance of objects from

| | |
|---|---|
| dist | distance, in which object switches to its next LoD, if d never turned off |

# Lighting, Smooth, TexMove, TexPing, Wobble - MAX

**Quad menu : properties / user defined**

## Lighting= ambient_mult, directional_mult, r, g, b

User Defined Properties:
Lighting=0.5,0.5,243,212,158

- all of the object's vertices have the same lighting
- this lighting is calculated as the multiple of ambient + multiple
- this is necessary to assign for Growtype objects

| | |
|---|---|
| ambient_mult | multiplier of sector's ambient light |
| directional_mult | multiplier of sector's directional light |
| r, g, b | color elements of light |

## Smooth= angle, group1, ..., groupN

User Defined Properties:
Smooth=83.0,1,21,30

- this is used for smoothing of shared edges of different objects
- the calculation is carried out when exporting into the *.bes format

| | |
|---|---|
| angle | angle for normal smoothing |
| group1, ... groupN | object groups |

## TexMove= tex_id, delta_u, delta_v

User Defined Properties:
TexMove= 1,0.012,0.017
TexMove= 2,-0.012,-0.017

- texture shift on the object
- can be set for individual textures

| | |
|---|---|
| tex_id | texture IDs:<br>0 - diffuse #1<br>1 - diffuse #2<br>2 - diffuse #3 |
| delta_u, delta_v | shift along u, v axes per second |

## TexPing= tex_id, u_time, u_min, u_max, u_soft, v_time, v_min, v_

- texture shift on the object: ping pong type
- can be set for individual textures

| | |
|---|---|
| tex_id | texture IDs:<br>0 - diffuse #1<br>1 - diffuse #2<br>2 - diffuse #3 |
| u_time, v_time | time of one shift there and back for u and v axes |
| u_min, u_max, v_min, v_max | shift interval for individual axes |
| u_soft, v_soft | direction change can be either smooth (curve) or hard (l<br>0 - 0 - hard change of direction<br>1 - soft change of direction |

## Wobble= AltX, AltY, AltX, SpeedX, SpeedY, SpeedZ

User Defined Properties:

Wobble=0.2,0.2,0.2,5,5,5

- this parameter makes it possibkle for objects to movearound its
- it is used mainly for plants
- recommended values can be for instance: 2.8, 3.1, 2.5 , 2.1, 1.6,

| | |
|---|---|
| AltX, AltY, AltZ | wobble deflection in degrees |
| SpeedX, SpeedY, SpeedZ | wobble speed |

# Ocluder, Portal, Sector - MAX

**Quad menu : properties / user defined**

## Ocluder= 1

User Defined Properties:

Ocluder=1

- an object with this parameter is an occluding object for portals
- for instance, it can be used inside a sector in the walls

## Portal= active_dist, half_dist, fade_dist, r, g, b

User Defined Properties:

Portal=30.0,15.0,10.0,0,4,79

| | |
|---|---|
| active_dist | distance from a portal to camera, to which it remain |
| half_dist | distance from a portal to camera, to which only those o had been linked to the sector in 3DS MAX |
| fade_dist | distance from a portal to camera, in which the portal to 0, the portal is closed without changing color |
| r, g, b | color of the closed portal |

## Sector= 1

User Defined Properties:

Sector=1

- an object with this parameter demarcates sector boundaries

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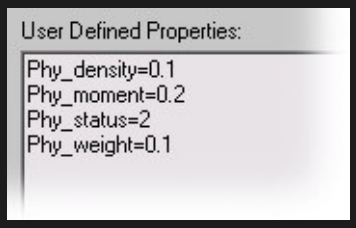
**Sub-entity - Ocluder**
**Sub-entity - Portal**
**Sub-entity - Sector**

# Physics - MAX

**Quad menu : properties / user defined**




For proper function of the physics module, it is necessary to fill

## Phy_density= val

- object density

## Phy_moment= val

- inertia momentum

## Phy_status= id

- the type of object's physical properties

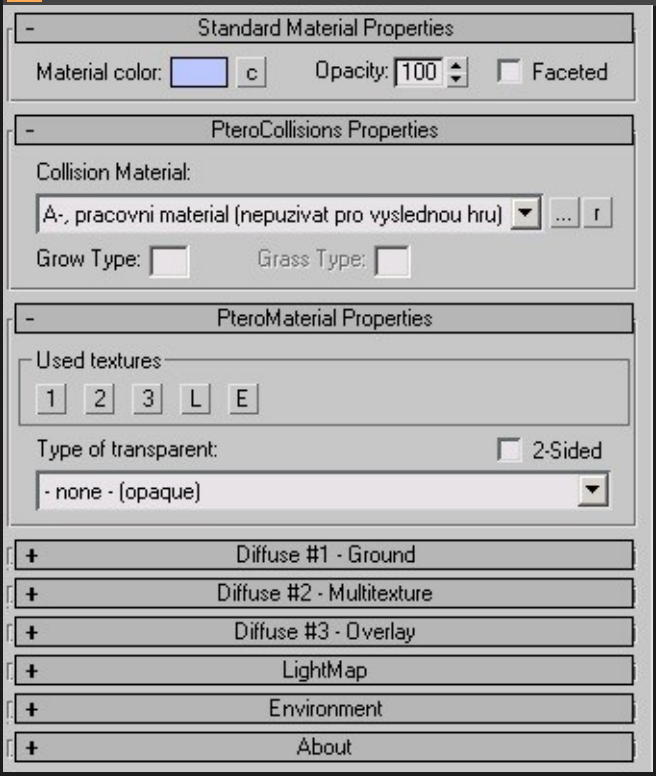
**id:**

| | |
|---|---|
| 0 | object collides, it is not synchronized in multiplayer |
| 1 | object collides, it is synchronized in multiplayer |
| 2 | object does not collide with physical objects, collides with p synchronized |
| 3 | object does not collide and it is not synchronized |

## Phy_weight= val

- object weight

# Using PteroMaterial




PteroMaterial is a plugin for 3DS MAX 5.1 which makes it c objects materials with all kinds of information for PteroEngine II. In prin extension of standard texturing material.

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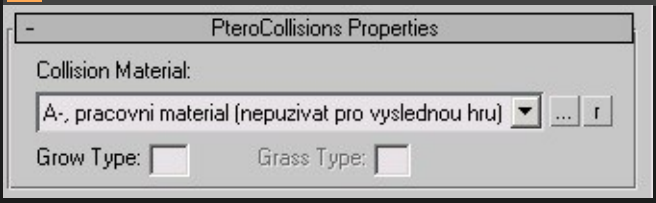
**Using texturing material**
**Using Collision Material**

# Using Collision Material




## PteroCollision - properties

1. **Collision material** - a library of materials used for collision typ etc.). The default material is set to A, but this should never
2. **Grow** - this is a one letter name of a directory in *\[g directory]\data\[sub-level directory]\grow*
3. **Typ trávy** - not supported yet!

| | |
|---|---|
| ... | opens the "browse" dialog for new path to the collision material |
| r | reloads database |

Using texturing material

# Using texturing material

## PteroMaterial - properties

1. **Used textures**
   - **1** - switches texture No.1 on/off; this is the base texture
   - **2** - switches texture No.2 on/off, this is the high-detail textur
   - **3** - switches texture No.3 on/off, this is an overlay textur
   - **L** - toggles Lightmap on/off
   - **M** - toggles Environment on/off

2. **Transparency types**
   - **žádná** - opaque
   - **typ 0** - transparent, stored in Z-buffer, sorted
   - **typ 1** - transparent, stored in Z-buffer, sorted, 2-bit Alpha
   - **typ 2** - translucent, NOT stored in Z-buffer, sorted
   - **typ 3** - transparent, stored in Z-buffer, NOT sorted, 2-bit A
   - **typ 4** - translucent, added to background, NOT stored

3. Using both sides for rendering

## Textures No. 1, 3 and lightmap

| | |
|---|---|
| E | toggles texture usage on/off |
| M | toggles distance-determined texture degradation (mipmaping) |
| U | toggles texture tiling along U axis |
| V | toggles texture tiling along V axis |
| Ch | mapping channel number (1-99) |
| i | cancels texture instance |
| u | cancels texture |
| r | reloads texture |
| s | toggles texture rendering on a model (on/off) |
| a | toggles texture's alpha channel rendering on a model (on/off) |
| t | toggles transparent texture rendering on a model (on/off) |

- note: the `s`, `a` and `t` buttons will work only if the "Show Map in Material Editor is on (

## Texture No.2

| | |
|---|---|
| E | toggles texture usage on/off |
| M | toggles distance-determined texture degradation (mipmaping) |
| U | toggles texture tiling along U axis |
| V | toggles texture tiling along V axis |
| T | tile count (1-999) |
| Ch | mapping channel number (1-99) |
| i | cancels texture instance |
| u | cancels texture |
| r | reloads texture |
| s | toggles texture rendering on a model (on/off) |
| a | toggles texture's alpha channel rendering on a model (on/off) |
| t | toggles transparent texture rendering on a model (on/off) |

- note: the `s`, `a` and `t` buttons will work only if the "Show Map in Material Editor is on (

## Environment map

| | |
|---|---|

| | |
|---|---|
| E | toggles texture usage on/off |
| M | toggles distance-determined texture degradation (mipmaping) |
| 1 | standard use |
| 2 | the map is applied according to its Alpha channel |
| Ch | mapping channel number (1-99) |
| i | cancels texture instance |
| u | cancels texture |
| r | reloads texture |
| s | toggles texture rendering on a model (on/off) |

- note: the `s` button will work only if the "Show Map in Viewport" bu
on ( )

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**Using Collision Material**
**Bitmap requirements**